Northeastern University

EECE 7205
Fundamentals of Computer Engineering

Greedy Algorithms with Arrays:
EV Charging Station

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Contents

The Problem **2**

Task 1: Understanding the Problem **2**

Task 2: Greedy Idea **3**

Task 3: Pseudocode **3**

Task 4: Hand Trace **3**

Task 5: Time Complexity **4**
- Frequency Count 4
- Final $T(n)$ 5
- Space Complexity 5
- Comparison with Checking Every Pair 5

Task 6: C++ Program **6**
- How the Heap Sort Works 6
- Program Output 6
- Test Results 6

Task 7: Think Further **7**

The Problem

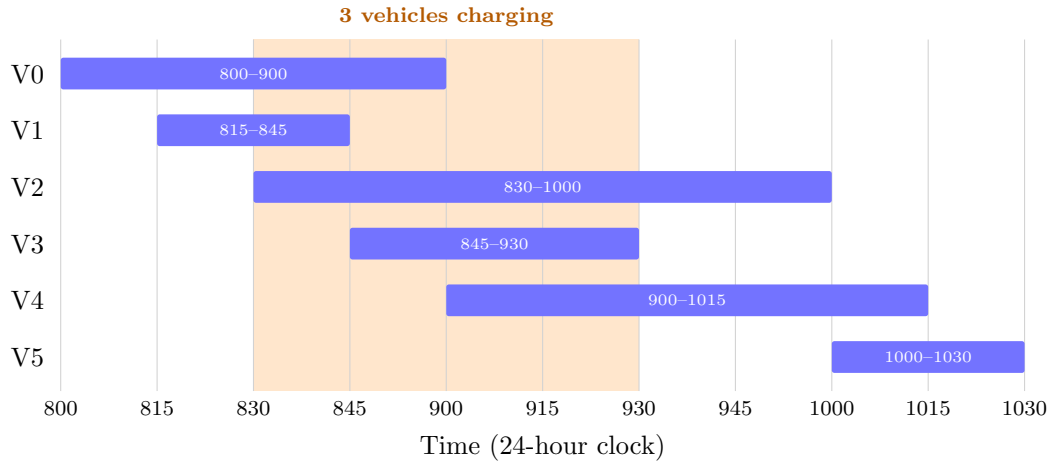
A campus EV charging station records each vehicle's arrival and departure time as a 24-hour integer (e.g. `830` means 8:30). Each vehicle needs its own charger from arrival until departure. We must find the smallest number of chargers so that no vehicle waits. A vehicle leaving at time t frees its charger at t, so a vehicle arriving at exactly t may use it.

Vehicle	0	1	2	3	4	5
`arrive`	800	815	830	845	900	1000
`depart`	900	845	1000	930	1015	1030

Expected output: `Chargers needed: 3`

Task 1: Understanding the Problem

Each vehicle is drawn as a bar from its arrival to its departure. The shaded band marks the period when the most vehicles are charging.



Counting the bars in each interval between consecutive events:

Interval	Vehicles charging	Change at start of interval	In use
800–815	V0	V0 arrives	1
815–830	V0, V1	V1 arrives	2
830–845	V0, V1, V2	V2 arrives	**3**
845–900	V0, V2, V3	V1 leaves, V3 arrives	**3**
900–930	V2, V3, V4	V0 leaves, V4 arrives	**3**
930–1000	V2, V4	V3 leaves	2
1000–1015	V4, V5	V2 leaves, V5 arrives	2
1015–1030	V5	V4 leaves	1

Answer: The most vehicles charging at the same time is **3**. This first happens at **8:30** (when vehicle 2 arrives, with vehicles 0 and 1 still charging) and the count stays at 3 from 830 until 930. At 845 and 900 one vehicle leaves at exactly the moment another arrives, so because of the equal-times rule the freed charger is reused and the count does not rise to 4. Therefore the station needs **3 chargers**.

Task 2: Greedy Idea

Sort the arrival times and the departure times separately, then sweep through all events in time order, always taking the earlier of the next arrival and the next departure (and taking the departure first when they are equal, so a freed charger can be reused immediately). An arrival *increases* the number of chargers in use by one, and a departure *decreases* it by one. The number in use at any moment equals the number of vehicles physically present, so the station needs at least the largest such value; and that many is also enough, because whenever a vehicle arrives the greedy choice is to hand it any charger that has been freed, so a new charger is only ever needed when the count rises above its previous maximum.

Task 3: Pseudocode

Algorithm 1 MinChargers($arrive[0..n-1]$, $depart[0..n-1]$, n)

```
1:  sort arrive in ascending order
2:  sort depart in ascending order
3:  i ← 0                                              ▷ index of next arrival
4:  j ← 0                                              ▷ index of next departure
5:  inUse ← 0,    most ← 0
6:  while i < n do
7:      if j < n and arrive[i] ≥ depart[j] then        ▷ departure first on a tie
8:          inUse ← inUse − 1                          ▷ charger freed
9:          j ← j + 1
10:     else
11:         inUse ← inUse + 1                          ▷ charger taken
12:         if inUse > most then
13:             most ← inUse
14:         i ← i + 1
15: return most
```

Handling equal times. The test uses $\geq$, not $>$. When $arrive[i] = depart[j]$ the departure is processed first, so the charger is released before the arriving vehicle takes one. Using $>$ instead would count both vehicles at once and overestimate the answer (it would give 4 on the sample data).

Why the loop can stop when $i = n$. After the last arrival only departures remain, and departures can only lower *inUse*, so *most* cannot change.

Task 4: Hand Trace

Sorted arrive: `800  815  830  845  900  1000`

Sorted depart: `845  900  930  1000  1015  1030`

"Next arrival" is $arrive[i]$ and "Next departure" is $depart[j]$ *before* the step is executed.

Step	Next arrival	Next departure	Event	Chargers in use	Most so far
1	800	845	Arrival ($800 < 845$)	1	1
2	815	845	Arrival ($815 < 845$)	2	2
3	830	845	Arrival ($830 < 845$)	3	3
4	845	845	Departure (tie, $845 \geq 845$)	2	3
5	845	900	Arrival ($845 < 900$)	3	3
6	900	900	Departure (tie, $900 \geq 900$)	2	3
7	900	930	Arrival ($900 < 930$)	3	3
8	1000	930	Departure ($1000 \geq 930$)	2	3
9	1000	1000	Departure (tie, $1000 \geq 1000$)	1	3
10	1000	1015	Arrival ($1000 < 1015$)	2	3
11	— ($i = n$)	1015	Departure (loop has ended)	1	3
12	— ($i = n$)	1030	Departure (loop has ended)	0	3

The algorithm exits after step 10 because all six arrivals have been processed. Steps 11 and 12 (in grey) show the two remaining departures for completeness; they only lower the count, so the maximum stays at 3.

Result: `Chargers needed: 3`, which matches the expected output and the peak found in Task 1.

Task 5: Time Complexity

Frequency Count

Let n be the number of vehicles. Each statement is given a constant cost c_k, and its frequency is the number of times it runs in the worst case. The sweep loop moves either i or j forward by one each pass, so it walks over both sorted arrays of size n.

#	Statement	Cost	Frequency	Total
1	`arr[i] = vehicles[i].first; (copy loop)`	c_1	n	$c_1 n$
2	`dept[i] = vehicles[i].second; (copy loop)`	c_2	n	$c_2 n$
3	`heapSort(arr, n);`	c_3	$n \log n$	$c_3\, n \log n$
4	`heapSort(dept, n);`	c_4	$n \log n$	$c_4\, n \log n$
5	`i=0; j=0; chargers=0; maxChargers=0;`	c_5	1	c_5
6	`while (i < n)`	c_6	$2n$	$2c_6 n$
7	`if (arr[i] >= dept[j])`	c_7	$2n - 1$	$c_7(2n - 1)$
8	`chargers--; j++;`	c_8	$n - 1$	$c_8(n - 1)$
9	`chargers++;`	c_9	n	$c_9 n$
10	`maxChargers = max(maxChargers, chargers);`	c_{10}	n	$c_{10} n$
11	`i++;`	c_{11}	n	$c_{11} n$
12	`return maxChargers;`	c_{12}	1	c_{12}

Where the frequencies come from.

- Lines 1–2 copy the n pairs into the two arrays `arr` and `dept`: $n + n = 2n$ steps.
- Lines 3–4 heap sort each array. Building the heap is $O(n)$ and the $n - 1$ extractions each sift down at most $\log n$ levels, so each sort is about $n \log n$ steps: $2n \log n$ in total.
- Lines 9–11 (an arrival) run exactly n times, once per vehicle.
- Line 8 (a departure) runs at most $n - 1$ times, because the loop ends as soon as the last arrival is handled.
- So the loop body runs at most $n + (n - 1) = 2n - 1$ times (line 7), and the loop test runs one extra time when it fails, $2n$ times (line 6). This is the pass over the two arrays of size n.

Final $T(n)$

Adding the Total column:

$$T(n) = (c_3 + c_4)\, n \log n + (c_1 + c_2 + 2c_6 + 2c_7 + c_8 + c_9 + c_{10} + c_{11})\, n + (c_5 + c_{12} - c_7 - c_8).$$

With every $c_k = 1$:

$$T(n) = 2n \log_2 n + 10n.$$

Grouping by part of the algorithm:

$$T(n) = \underbrace{O(n \log n)}_{\text{two sorts}} + \underbrace{O(2n)}_{\text{pass over 2 arrays of size } n}.$$

Big-O drops constant factors, so $O(2n) = O(n)$, and since $n \log n$ grows faster than n, the sorting term dominates:

$$\boxed{T(n) = O(n \log n) + O(2n) = O(n \log n)}$$

Space Complexity

The function stores two arrays of n integers (`arr` and `dept`) plus four scalar variables (`i`, `j`, `chargers`, `maxChargers`):

$$S(n) = n + n + 4 = 2n + 4 \quad \Longrightarrow \quad \boxed{S(n) = O(2n) = O(n)}$$

The input `vehicles` is passed by `const` reference, so it is not copied. Heap sort works in place and needs only $O(1)$ extra memory (a few index variables), so it does not change the result.

Comparison with Checking Every Pair

The simple method takes each vehicle k and compares it with every other vehicle to count how many are charging at the moment $arrive[k]$ (the peak always begins at some arrival). That is n checks for each of n vehicles, so it costs $\boldsymbol{O(n^2)}$. For $n = 100$ this is about $100^2 = 10{,}000$ comparisons, while the greedy method needs roughly $2 \cdot 100 \log_2 100 + 200 \approx 1{,}500$. The gap grows quickly with n: for $n = 10{,}000$ it is about 10^8 versus 3×10^5.

Task 6: C++ Program

The full source code is available on GitHub:

`https://github.com/roh9031/EECE---7205-Class-Assignment-`

The implementation follows the pseudocode from Task 3. The arrival and departure times are passed in as {`arrive, depart`} pairs, which are split into two arrays, `arr` and `dept`. Each array is sorted with a hand-written **heap sort**, and then the greedy sweep counts the chargers in use. The vehicle list is passed by `const` reference, so it is not copied and cannot be changed by the function.

How the Heap Sort Works

`siftDown` moves a value down a max-heap: it compares the value with its larger child and swaps them until the value is at least as large as both children. `heapSort` first builds a max-heap by sifting down every non-leaf node, from index $n/2 - 1$ back to 0. It then repeatedly swaps the largest value (the root) to the end of the array and sifts the new root down over the remaining, shorter heap. Building the heap costs $O(n)$ and the $n - 1$ extractions cost $O(\log n)$ each, so heap sort runs in $O(n \log n)$ time and sorts the array in place with $O(1)$ extra memory.

Program Output

```
Test case: all arrive together - {900,1000},{900,1000},{900,1000},{900,1000}
Chargers needed: 4

Test case: all arrive back to back - {900,1000},{1000,1100},{1100,1200}
Chargers needed: 1

Test case: random timings -
    {800,900},{815,845},{830,1000},{845,930},{900,1015},{1000,1030}
Chargers needed: 3
```

Test Results

Test	Input {arrive, depart}	Expected output	My output	Pass?
Sample (random timings)	{800,900}, {815,845}, {830,1000}, {845,930}, {900,1015}, {1000,1030}	`Chargers needed: 3`	`Chargers needed: 3`	**Yes**
All arrive together	{900,1000} × 4	`Chargers needed: 4`	`Chargers needed: 4`	**Yes**
Back-to-back	{900,1000}, {1000,1100}, {1100,1200}	`Chargers needed: 1`	`Chargers needed: 1`	**Yes**

The back-to-back case confirms the equal-times rule: each vehicle leaves at the exact moment the next one arrives, so a single charger is reused all morning.

Task 7: Think Further

After sorting, $arrive[k]$ and $depart[k]$ usually belong to different vehicles, but the algorithm never needs to know which departure goes with which arrival. At any moment t, the number of chargers in use is simply

$$(\text{number of arrivals up to } t) \;-\; (\text{number of departures up to } t),$$

with departures at exactly t counted first. Both counts depend only on the *set* of arrival times and the *set* of departure times, not on how they are paired. Sorting each array keeps every time value and only changes the order, so these counts, and therefore their maximum, are unchanged.

In other words, a charger is not tied to a particular vehicle: when any vehicle leaves, its charger becomes available to whoever arrives next. The sweep tracks only how many chargers are busy, which is exactly what the question asks.

The count also never goes negative. Since every vehicle departs after it arrives, the k-th smallest departure time is always later than the k-th smallest arrival time. So the algorithm can never process more departures than arrivals, and $inUse$ always describes a real situation.